SHAP Stability Index as a Leading Indicator of Concept Drift in Credit Risk Gradient Boosting Models

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Abstract

Credit risk and fraud models are usually calibrated on historical data, but production portfolios evolve as borrowers, fraud strategies, and product rules change. This temporal mismatch can weaken a deployed model before labelled outcomes are available for validation. Existing monitoring practice mainly tracks score distributions, input distributions, or realised error rates; it does not directly ask whether the model's explanation profile has changed. This paper defines the SHAP Stability Index (SSI), which measures the consistency of global SHAP feature-importance rankings between adjacent evaluation windows. We evaluate XGBoost, LightGBM, and CatBoost in a streaming temporal protocol with ADWIN-triggered adaptive retraining on IEEE-CIS Fraud Detection (590,540 transactions, six months), ULB Credit Card Fraud (284,807 transactions, two days), and Give Me Some Credit (150,000 records, used as a distributional stability control). On IEEE-CIS, across 16-17 ADWIN-detected drift events, SSI falls 7.8 evaluation windows on average (about **55** calendar days) before AUC degradation is observable (Wilcoxon signed-rank: $\boldsymbol{p<0.001}$, $\boldsymbol{r=0.88}$ for all three architectures). SSI also separates the observed regimes: **0.66-0.72** under persistent drift, **0.95-0.99** under localised drift, and **>=0.987** on the stability control, where it raises no spurious alerts. SSI is therefore a label-free explanation-stability monitor that complements performance and distributional checks in credit model governance.

Keywords: concept drift, credit risk, SHAP, model monitoring, gradient boosting, feature importance stability

1 Introduction

Machine learning classifiers now support decisions in domains ranging from medical image diagnosis [1] to financial services. In credit risk and fraud detection, the stakes are especially direct: underwriting models shape access to credit, pricing, expected loss, and fraud intervention. The Federal Reserve's Model Risk Management guidance (SR 11-7) [2] therefore requires financial institutions to validate and monitor quantitative models used for default probability, creditworthiness, and fraud likelihood. For structured credit data, gradient boosting methods such as XGBoost [3] and LightGBM [4] are widely used because they perform strongly against logistic regression and neural network baselines [5, 6].

Deployment changes the problem. A model is fitted to an earlier sample, while future borrowers, merchants, fraud tactics, and product policies continue to move. These changes alter $P(\mathbf{x}, y)$ over time, the setting formalised as concept drift [7]. The operational risk is not a software failure but a monitoring failure: predictions can remain available while their statistical reliability declines. Surveys of financial ML deployments identify temporal model decay as a major unresolved production risk [6, 8].

Credit portfolios also suffer from delayed labels. Defaults, charge-offs, and confirmed fraud outcomes can arrive 30-90 days after the original decision, so the loss function cannot be measured at the time the decision is made. Sculley et al. [9] discuss this delay as a source of technical debt in deployed ML systems, and Grzenda et al. [10] show that detection lag for error-based methods increases with the label-delay window. As a result, detectors such as ADWIN [11], DDM [12], and EDDM [13] cannot fire until outcome feedback exists. Label-free alternatives address only part of the problem. Margin Density Drift Detection tracks decision-boundary density [14], while the Population Stability Index (PSI) [15] tracks marginal score or feature distributions. Neither directly measures whether the fitted model has changed its feature reliance while producing apparently stable scores.

A second driver of urgency comes from the regulatory evolution of interpretability requirements. The European Union Artificial Intelligence Act [16] classifies credit scoring as a high-risk AI application subject to mandatory transparency, traceability, and human oversight requirements throughout the entire operational lifetime of the model, not just at initial validation. Bücker et al. [17] showed empirically that the explanatory quality of deployed credit scoring models depends critically on whether *explanation consistency* is monitored over time rather than assessed only at deployment. This creates a dual obligation for credit institutions: they must detect distributional change early enough to protect predictive performance, and they must be able to demonstrate that the model's feature-level decision rationale has not shifted in ways that would compromise regulatory traceability. Neither obligation is satisfied by existing monitoring practice.

Explanations provide a second monitoring surface. SHAP (SHapley Additive exPlanations) [18] assigns feature-level contributions to individual predictions, and TreeSHAP [19] computes those attributions efficiently for tree ensembles. Aggregating absolute SHAP values over an evaluation window gives a ranked summary of the features driving a model's decisions for that cohort. Stable rankings imply stable feature reliance; sharp rank movement implies that the model is using a different explanatory structure on recent data. Lin and Wang [20] study SHAP ranking stability in a static credit default model under resampling, but they do not test whether ranking movement accumulates in a temporal stream or whether it precedes later performance degradation.

We answer these questions by defining the **SHAP Stability Index (SSI)**, computed as the Spearman rank correlation between global SHAP feature importance vectors in adjacent evaluation windows. When SSI falls, the model is not merely changing its score distribution; it is changing which variables carry the decision. That signal is available at scoring time because it uses the current feature vectors and the fitted model, rather than delayed outcome labels. SSI therefore fills a monitoring gap left by PSI, error-based drift detectors, and one-off explanation reviews: it tracks whether the deployed model's feature reliance remains stable as new data arrive. We test this signal with XGBoost, LightGBM, and CatBoost on three public credit datasets, using ADWIN-triggered adaptive retraining and Optuna-based hyperparameter optimisation [21].

The study contributes the following:

1. A temporal evaluation design for credit drift experiments that combines three gradient boosting architectures (XGBoost, LightGBM, CatBoost), ADWIN drift detection, and Optuna-tuned adaptive retraining across three structurally distinct public datasets: IEEE-CIS Fraud Detection (590,540 transactions, six months), ULB Credit Card Fraud (284,807 transactions, two days), and Give Me Some Credit (150,000 records, used as a distributional stability control).
2. The SHAP Stability Index (SSI), a label-free statistic for comparing global SHAP importance rankings across successive evaluation windows without waiting for ground-truth outcomes.
3. Evidence that SSI acts as a *leading indicator* on the IEEE-CIS stream: across 16-17 ADWIN-detected drift events, SSI declined an average of 7.8 evaluation windows (about 55 calendar days) before AUC deterioration became detectable, with the same pattern across all three model architectures (Wilcoxon $p<0.001$, $r=0.88$).
4. Cross-dataset validation showing that the same SSI scale separates three regimes: 0.66-0.72 under persistent drift (IEEE-CIS), 0.95-0.99 under localised drift (ULB CC Fraud), and >=0.987 on a distributional stability control (GMSC), without threshold recalibration by model family.

The remainder of this paper is organised as follows. Section 2 reviews related work on gradient boosting for credit risk, concept drift detection, and SHAP in financial governance. Section 3 formalises the problem and presents the SSI methodology. Section 4 describes the experimental setup. Section 5 presents and discusses the results. Section 6 concludes with limitations and future directions.

2 Related Work

2.1 Gradient Boosting for Credit Risk and Fraud Detection

Gradient boosting is the main tree-ensemble family used for tabular credit risk modelling. Chen and Guestrin [3] introduced XGBoost, which combines regularised boosting with column subsampling, shrinkage, and cache-efficient split enumeration. Ke et al. [4] introduced LightGBM, replacing depth-wise tree growth with leaf-wise growth and using gradient-based one-side sampling to reduce the cost of fitting large datasets. Prokhorenkova et al. [22] introduced CatBoost, which handles categorical variables through ordered target statistics and symmetric oblivious trees, reducing target leakage from conventional encoding. These systems cover the practical design space most relevant to credit risk: training speed, memory use, and categorical feature handling.

Credit-scoring benchmarks give a consistent reason to focus on these models. Lessmann et al. [5] compared 41 classifiers on eight public credit datasets and reported that gradient boosting outperformed logistic regression, neural networks, and support vector machines on both accuracy and expected-profit measures. Dumitrescu et al. [23] showed that non-linear effects learned by gradient boosting can be incorporated into logistic regression scorecards, improving accuracy without abandoning interpretable scorecard structure. In a review of 76 machine-learning credit-risk studies, Shi et al. [6] likewise identified gradient boosting as a leading method, while noting that temporal decay and distribution shift remain open deployment problems.

Credit card fraud detection presents a closely related modelling challenge. Bhattacharyya et al. [24] demonstrated the superiority of ensemble methods for fraud classification under severe class imbalance, a challenge typically addressed through synthetic oversampling [25] or cost-sensitive weighting. Bahnsen et al. [26] showed that temporal and velocity features computed over transaction history are a primary performance driver beyond point-in-time attributes. Lebichot et al. [27] demonstrated that incremental learning strategies can match batch retraining performance in production fraud detection pipelines operating under concept drift, providing empirical evidence that adaptive modelling is necessary in non-stationary transaction streams.

Broader tabular benchmarks support the same modelling choice. Shwartz-Ziv and Armon [28] evaluated 48 datasets and found that tree-based ensembles matched or exceeded neural networks on most tasks, especially when continuous and categorical variables appear together. Across 45 datasets, Grinsztajn et al. [29] linked this advantage to irregular feature geometry, a condition under which neural networks still struggle even after architecture search. Gorishniy et al. [30] reported that transformer architectures for tabular data do not reliably surpass well-tuned gradient boosting on mixed feature sets. McElfresh et al. [31], benchmarking 176 datasets and 19 algorithms, found the strongest GBDT advantage on heavy-tailed or irregular feature distributions, which are common in transaction data.

Beyond accuracy, gradient boosting has attracted scrutiny on fairness and transparency. Kozodoi et al. [32] demonstrated that fairness-aware models can achieve competitive AUC while measurably reducing discriminatory outcomes, at the cost of a quantifiable profit trade-off. Bücker et al. [17] concluded that the explanatory quality of deployed credit scoring models depends critically on whether explanation consistency is monitored over time rather than assessed only at initial validation - a finding that directly motivates the SSI framework.

These benchmarks are informative but mostly static. They do not test how the same models behave after deployment, when the evaluation stream keeps moving and the relationship between features and outcomes can change. The experiments in Section 4 are designed around that temporal setting.

2.2 Concept Drift Detection in Data Streams

Concept drift refers to changes in the joint distribution $P(\mathbf{x}, y)$ over time that reduce the predictive validity of a trained model. Gama et al. [7] provided the foundational taxonomy, distinguishing gradual, abrupt, incremental, and recurring drift together with a unified review of detection and adaptation strategies. Webb et al. [33] refined this taxonomy by separating virtual drift, in which only the marginal $P(\mathbf{x})$ shifts, from real drift, in which the posterior $P(y \mid \mathbf{x})$ changes. The distinction has practical consequences: virtual drift may not require retraining, whereas real drift requires model adaptation to maintain predictive validity.

Statistical drift detectors.

Performance-based detectors watch an online error stream and declare drift when recent errors no longer look consistent with earlier errors. ADWIN (Adaptive WINdowing) [11] keeps a variable-length window and cuts it when two subwindows differ enough to satisfy its statistical test. DDM [12] instead compares the current error process with its best historical point, while EDDM [13] shifts the focus from error rate to the spacing between errors to improve sensitivity under gradual drift. Other detectors monitor different signals: KSWIN [34] applies a Kolmogorov-Smirnov comparison to recent univariate observations, HDDM [35] uses Hoeffding-test warning and drift levels, and the Page-Hinkley test [36] accumulates directional deviations in a scalar sequence.

These methods are useful but not interchangeable. They differ in how quickly they react, how often they false alarm, and whether labels are required. We use ADWIN because its adaptive window fits the bursty drift pattern expected in transaction streams and because River [37] provides a stable implementation that can be embedded directly into the evaluation loop.

Adaptive learning and monitoring frameworks.

Street and Kim [38] proposed ensemble-based streaming classifiers that retire stale component models and train new ones at block boundaries, implicitly handling gradual drift. Losing et al. [39] reviewed incremental online learning methods and found that detector-triggered full retraining outperforms passive forgetting under abrupt drift - precisely the regime ADWIN is designed to identify. Klaise et al. [40] formalised the case for integrating drift detection, outlier detection, and explanation monitoring as complementary components of a single production observability pipeline, noting that no single signal is sufficient in isolation.

Surveys and the detection gap.

Bayram et al. [41] reviewed performance-aware drift detectors and confirmed that all current methods measure model degradation that has already manifested in prediction errors rather than anticipating it from upstream signals. The 2024 two-part survey by Hinder et al. addresses both the detection question [8] and the drift localisation and explanation question [42], collectively identifying the temporal gap between drift onset and detector response as the central open problem in the field. Lu et al. [43] reviewed learning-under-drift approaches across domains and noted that interpretability of drift causes remains largely absent from the literature.

Production monitoring in financial machine learning.

In credit model monitoring, the Population Stability Index (PSI) is commonly used to compare a reference population with a later monitoring population [15]. It bins a score or feature into n intervals and computes:

$$\mathrm{PSI} = \sum_{i=1}^{n} \big(A_i - E_i\big) \ln\!\Big(\tfrac{A_i}{E_i}\Big), \tag{1}$$

where A_i and E_i denote monitoring and reference proportions in bin i. Values above 0.25 are typically treated as requiring review under SR 11-7 [2]. The Characteristic Stability Index (CSI) applies the same calculation feature by feature. PSI and CSI are therefore well suited to distribution checks, but their unit of analysis is still a marginal distribution. They do not say whether the fitted model is relying on the same features for its decisions. SSI is introduced for that missing internal-logic signal.

The label delay problem.

In production credit systems, labels often arrive after the monitoring decision is needed. Sculley et al. [9] describe label delay and model decay as technical debt in deployed ML systems. Sethi and Kantardzic [14] avoid immediate label dependence by monitoring the density of observations near the classifier margin, and Grzenda et al. [10] show that error-based drift detection slows as the delay window grows. SSI follows the same operational constraint: it uses the current feature vectors and fitted model at inference time, not resolved outcomes.

2.3 Explainable AI in Credit Risk Governance

Interpretability in consumer credit is now part of model governance, not only a post-hoc reporting preference. SR 11-7 [2] requires ongoing model validation and monitoring, while the Basel Committee's operational risk principles [44] impose related transparency expectations for models used in capital contexts. The European Union Artificial Intelligence Act [16] treats credit scoring as a high-risk AI use case, bringing traceability and human oversight into the operating life of the model. If a model's SHAP importance rankings move sharply over time, the explanation record available to auditors has changed even when aggregate AUC remains acceptable.

Two post-hoc explanation methods are especially common in financial ML. LIME [45] fits a local surrogate around an individual prediction. SHAP [18] assigns Shapley-value attributions to features, with efficiency, symmetry, dummy, and linearity properties. Subsequent work shows why these explanations require care: Sundararajan and Najmi [46] analyse how the Shapley-value operationalisation and background distribution affect attributions, while Covert et al. [47] place SHAP, LIME, and permutation importance in a shared removal-based framework. TreeSHAP [19] is the relevant variant here because it computes exact attributions for tree ensembles efficiently. Arrieta et al. [48] identify temporal consistency as an open XAI issue, but most stability studies still evaluate one trained model under perturbations rather than a model deployed through a changing stream.

The stability and fragility of post-hoc explanations have been examined from several angles with direct governance implications. Alvarez-Melis and Jaakkola [49] defined a self-consistency criterion requiring that similar inputs produce similar explanations and showed that neither LIME nor SHAP satisfies it uniformly, establishing precedent for treating explanation stability as a measurable governance property. Slack et al. [50] demonstrated that both methods can be adversarially manipulated to conceal discriminatory behaviour behind innocuous-looking explanations, underscoring that static explanation audits are insufficient for deployed systems under distribution shift. Fryer et al. [51] showed that Shapley values for feature selection can produce misleading importance rankings under correlated features - a common condition in credit data - indicating that attribution instability is a concern even before distributional shift is introduced. Molnar et al. [52] catalogued general pitfalls of model-agnostic interpretation methods, including failure to account for feature dependence and production of spurious attributions in high-dimensional settings, concluding that one-time explanation audits provide insufficient governance coverage for production systems.

In the credit domain specifically, Bussmann et al. [53] integrated SHAP attribution monitoring into an end-to-end credit risk management framework, treating SHAP values as first-class governance artefacts alongside AUC and Gini coefficient and demonstrating the practical utility of feature-level explanation tracking in production. Guidotti et al. [54] identified explanation stability as a key desideratum alongside fidelity, observing that explanations which change arbitrarily between consecutive decision windows provide no actionable governance signal. Molnar [55] discusses global feature importance consistency as a practical property analysts should assess but provides no quantitative framework for monitoring it over time.

The most directly related prior work is Lin and Wang [20], who examined SHAP ranking stability in a static credit card default model across resampling strategies and class imbalance conditions, finding significant rank variation across experimental configurations. Their study is restricted to a single cross-sectional model under controlled perturbations and does not examine whether SHAP rankings shift systematically across temporal evaluation windows as concept drift accumulates, nor whether such shifts carry predictive information about upcoming performance degradation before it surfaces in outcome metrics. These are precisely the questions addressed by the SHAP Stability Index introduced in Section 3.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{D} = \{(\mathbf{x}_t, y_t)\}_{t=1}^{T}$ denote a temporal stream of credit transactions, where $\mathbf{x}_t \,\text{in}\, \mathbb{R}^d$ is the feature vector at time t and $y_t \,\text{in}\, \{0, 1\}$ is the binary outcome label. We partition $\mathcal{D}$ into K non-overlapping evaluation windows $W_1, W_2, \ldots, W_K$ of fixed size w with stride s, preserving temporal order so that no future observations enter any window's training set.

At each window k, a gradient boosting model $\mathcal{M}_k$ is trained on all data preceding W_k and evaluated on W_k. Two regimes are compared throughout this study:

- **Static**: $\mathcal{M}_k = \mathcal{M}_1$ for all k; the model is trained once at initialisation and never updated after deployment.
- **Adaptive**: $\mathcal{M}_k$ is retrained whenever an online drift detector signals a statistically significant change in the model's error rate.

A concept drift event occurs at window k when the joint distribution $P_k(\mathbf{x}, y)$ differs significantly from $P_{k-1}(\mathbf{x}, y)$ as determined by ADWIN [11]. Real drift, in which the conditional $P_k(y \mid \mathbf{x})$ shifts, degrades the static model's predictions systematically, whereas the adaptive model recovers by retraining on the most recent data.

3.2 The SHAP Stability Index

At each evaluation window k, TreeSHAP [19] is applied to a stratified subsample of up to 2,000 instances drawn from W_k. TreeSHAP computes exact Shapley values for tree ensembles in polynomial time without the exponential enumeration required by the general Shapley formula, making it tractable for high-dimensional credit datasets evaluated repeatedly across many windows.

The global feature importance vector $\mathbf{s}_k \,\text{in}\, \mathbb{R}^d$ is defined component-wise as the mean absolute SHAP value across the sampled instances:

$$s_k^{(j)} = \frac{1}{|S_k|} \sum_{i \mathrm{in} S_k} \big| s_j(\mathbf{x}_i) \big|, \qquad j = 1, \ldots, d, \tag{2}$$

where S_k inside W_k is the evaluation sample and $s_j(\mathbf{x}_i)$ is the SHAP attribution for feature j on instance $\mathbf{x}_i$. Let $\mathbf{R}_k \,\text{in}\, \mathbb{Z}^{d'}$ denote the rank vector obtained by ordering the top $d' = 20$ components of $\mathbf{s}_k$ in descending order. Restricting to the top-d' features reduces noise from low-importance features whose ranks fluctuate arbitrarily near the tail of the distribution.

Definition 1 (SHAP Stability Index) The **SHAP Stability Index** at window k with lookback L is:

$$\mathrm{SSI}(k) = \frac{1}{L} \sum_{h=1}^{L} r_s(\mathbf{R}_{k-h+1}, \, \mathbf{R}_{k-h}) \,, \tag{3}$$

where r_s denotes the Spearman rank correlation coefficient.

Spearman correlation is used in preference to Pearson correlation because SHAP magnitudes vary in scale across windows independently of rank structure; the ordinal measure captures genuine reordering while remaining invariant to monotone rescaling of attribution values. $\mathrm{SSI}(k) = 1$ indicates an identical feature importance ordering to the preceding window; values near zero indicate a near-random rearrangement. SSI requires only model outputs and unlabelled feature vectors and can therefore be computed at inference time before ground-truth labels become available.

SSI as a leading indicator.

For each ADWIN-detected drift event, we measure the lead time L_{lead} of SSI relative to AUC degradation by defining three quantities:

- k_{SSI}: the first window at which $\mathrm{SSI}(k) < T_{\text{SSI}}$, where the instability threshold $T_{\text{SSI}} = 0.80$.
- k_{AUC}: the first window after drift onset at which the AUC falls more than $D_{\text{AUC}} = 0.02$ below the pre-drift baseline.
- Lead time: $L_{\text{lead}} = k_{\text{AUC}} - k_{\text{SSI}}$ (in evaluation windows). A positive value indicates that SSI signalled instability before AUC degradation was detectable.

Parameter selection rationale.

The SSI hyperparameters d', L, and T_{SSI} are motivated as follows.

Top-d' truncation ($d' = 20$). In gradient boosting models for tabular credit data, feature importance follows a heavy-tailed distribution: a small number of features accounts for the majority of total mean absolute SHAP attribution, while the tail consists of near-zero contributions whose rank order fluctuates arbitrarily from window to window. Restricting SSI to the top $d' = 20$ features by mean absolute SHAP value focuses the rank correlation on the semantically meaningful portion of the importance ordering and removes the noise introduced by features operating near zero attribution. The value $d' = 20$ is chosen to exceed the approximate "elbow" in the SHAP importance distribution observed during exploratory analysis of the IEEE-CIS dataset, beyond which additional features contribute negligible marginal attribution.

Lookback window ($L = 5$). The lookback L determines the temporal span of the rolling average in Equation (3). With the 7-day evaluation stride on IEEE-CIS, $L = 5$ yields a 35-day effective monitoring window, matching the sub-monthly review horizon at which model risk governance processes are typically active in credit institutions. Values $L < 3$ produce a volatile one-shot measure sensitive to single-window noise; values $L > 7$ introduce excessive lag that would delay detection of abrupt drift episodes. $L = 5$ balances responsiveness with noise suppression.

Instability threshold ($T_{\text{SSI}} = 0.80$). The threshold determines which SSI values are classified as indicating feature-ranking instability. A sensitivity analysis reported in Section 5.5 demonstrates that SSI's first crossing below threshold is invariant to T over the range $[0.66, 0.90]$ for both model families on the IEEE-CIS dataset: all values in this range detect the onset of instability at the same evaluation window. Values below 0.65 reduce coverage to fewer than 30% of the drift-affected windows. The chosen $T_{\text{SSI}} = 0.80$ lies in the middle of the stable detection range and is consistent

Algorithm 1 Temporal Streaming Evaluation with ADWIN and SSI Monitoring

Require: Windows $W_1, \ldots, W_K$; model $\mathcal{M}$; ADWIN(d_{ADWIN}); lookback L; buffer size B

Train $\mathcal{M}$ on windows $W_1, \ldots, W_{\text{init}}$
Initialise training buffer $\mathcal{B} \leftarrow \{W_1, \ldots, W_{\text{init}}\}$
for $k = \text{init} + 1$ **to** K **do**
  Compute AUC_k, error_k on W_k using $\mathcal{M}$
  Feed error_k to ADWIN(d_{ADWIN})
  Compute $\mathbf{s}_k$ via TreeSHAP on stratified sample of W_k
  Compute SSI(k) via Equation (3)
  if ADWIN signals drift **then**
    Retrain $\mathcal{M}$ on the B most recent windows in $\mathcal{B}$
    Log: k, AUC_k, SSI(k), lead time L_{lead}
  end if
  Add W_k to $\mathcal{B}$; drop oldest window if $|\mathcal{B}| > B$
end for
return $\{\text{AUC}_k\}_k$, $\{\text{SSI}(k)\}_k$, drift event log

with Spearman correlation thresholds used in analogous rank stability audits in the psychometric literature.

3.3 ADWIN-Triggered Adaptive Retraining

ADWIN maintains a self-adjusting window W over a scalar performance signal, specifically the per-window classification error rate $(1 - \text{accuracy})$. At each step, ADWIN tests whether the mean of any contiguous sub-window W_0 inside W differs significantly from the mean of the complement $W_1 = W \setminus W_0$:

$$\left| m_{W_0} - m_{W_1} \right| >= e_{\text{cut}}, \tag{4}$$

where e_{cut} is derived from the Hoeffding bound at confidence level $1 - d_{\text{ADWIN}}$ [11]. The window is trimmed from its oldest end until no such partition satisfies Equation (4). When a cut is found, a drift event is declared and the adaptive model is retrained immediately. The ADWIN implementation is provided by the River streaming machine learning library [37], which is consistent with the original algorithm specification [11].

Algorithm 1 describes the complete streaming evaluation loop applied to every dataset and model combination in this study.

3.4 Hyperparameter Optimisation

XGBoost, LightGBM, and CatBoost are each tuned using Optuna [21] with the Tree-structured Parzen Estimator (TPE) sampler [56]. TPE models the distribution of hyperparameters that produced high-performing trials as one Gaussian mixture density $p_{\text{good}}(z)$, and that of low-performing trials as another, $p_{\text{bad}}(z)$, then samples candidates where the ratio $p_{\text{good}}(z)/p_{\text{bad}}(z)$ is maximised. This concentrates search effort in promising regions without requiring a global surrogate regression model.

The tuning objective is AUC evaluated on a held-out temporal validation window W_{val} placed immediately after the training period, ensuring that no future data influences hyperparameter selection. Each tuning run executes 150 trials subject to a two-hour wall-clock timeout. The Median pruner terminates unpromising trials whose intermediate AUC on W_{val} falls below the median of completed trials at the same optimisation step. All optimised hyperparameter sets are serialised as JSON files for exact reproducibility. The full parameter search spaces for each model are listed in Section 4.

4 Experimental Setup

4.1 Datasets

Three publicly available datasets spanning distinct credit risk and fraud detection tasks are used to evaluate the proposed framework. Table 1 summarises their key characteristics.

IEEE-CIS Fraud Detection. This dataset contains 590,540 card transactions provided as two source files, transaction attributes and identity metadata, joined on a shared transaction identifier. The binary label `isFraud` is imbalanced at approximately 3.5% positive. The merged raw feature space exceeds 400 columns, comprising the transaction amount, 14 count features (`C1-C14`), 15 time-delta features (`D1-D15`), nine binary match flags (`M1-M9`), 339 anonymised `V`-features, card identifiers, and device identity attributes. A real calendar timestamp (`TransactionDT`, in seconds relative to a reference date) enables genuine temporal ordering across a six-month collection period.

Give Me Some Credit (GMSC) - distributional stability control. This dataset contains 150,000 borrower records with binary label `SeriousDlqin2yrs` (about 6.7% positive) and 10 credit-bureau features. It contains no real timestamp. A pseudo-temporal ordering is constructed by sorting on age (ascending) then revolving utilisation (descending), which creates a smooth demographic gradient rather than a genuine temporal sequence. GMSC is included exclusively as a *distributional stability control* (negative control): it tests whether SSI produces spurious low-stability alerts when data are ordered by demographic attributes rather than time. Any observation of high SSI (>=0.987) and zero ADWIN events on GMSC is expected by design and cannot be cited as drift-detection validation; it confirms only that the metric does not false-positive on smoothly demographically ordered data.

ULB Credit Card Fraud - localised-drift control. This well-studied benchmark [57] contains 284,807 European card transactions recorded over two calendar days in September 2013. The binary label `Class` indicates fraud (492 events, 0.17%). Features `V1-V28` are principal components of anonymised transaction attributes; only `Time` (elapsed seconds since the first transaction) and `Amount` are provided in raw form. Its two-day span yields a single ADWIN drift event, making it a localised-drift control rather than a persistent-drift experiment.

Table 1 Summary of the three evaluation datasets. Positive rate is the fraction of fraudulent or delinquent instances. Engineered feature counts reflect the final dimensionality after all transforms and selection steps described in Section 4.2. GMSC has no real timestamp and is used as a distributional stability control only.

Dataset	Records	Raw	Eng.	Pos. (%)	Span	Temporal Basis
IEEE-CIS Fraud	590,540	>400	200	3.5	6 mo.	TransactionDT
GMSC	150,000	10	17	6.7	N/A	Stability control
ULB CC Fraud	284,807	30	38	0.17	2 d.	Time (s)

4.2 Feature Engineering

All feature transforms are fitted exclusively on data preceding each evaluation window and applied without re-fitting to that window, eliminating any look-ahead leakage across the temporal boundary.

IEEE-CIS. `TransactionAmt` yields three derived predictors: log-transform, decimal-part extraction, and a binary round-amount flag. Calendar features (day of week, hour of day, weekend indicator) are decoded from the `TransactionDT` timestamp. Transaction amounts are aggregated within seven cardholder-group columns (`card1-card6`, purchaser email domain) to produce per-group count, mean, standard deviation, and maximum statistics. Each of the 15 `D`-columns receives a log-transform and a binary missing-value indicator. Highly correlated `V`-feature pairs (Pearson $r > 0.95$ on training data) are removed to eliminate multicollinearity. Categorical columns, including product code, card network, email domain, match flags, and device type, are target-encoded with Laplace smoothing (prior weight $k = 300$). Identity fields (`id_01-id_38`) are label-encoded with an explicit *unknown* token to handle categories unseen in evaluation windows. Residual missing numerical values are imputed with -999. A LightGBM relevance filter retains the top 200 features by training-set Gini importance, yielding a final 200-dimensional representation.

ULB CC Fraud. `Amount` is log-transformed, z-scored using training-partition statistics, and a round-amount indicator is added. The `Time` field produces hour-of-day, day-of-period, and a night-hours indicator (00:00-06:00). Five pairwise products among the first four principal components (`V1-V4`) are appended as interaction features, yielding 38 dimensions in total.

GMSC. Age values of zero are replaced by the training-partition median. Delinquency counts are capped at 90. Four derived features are added: total late payments, debt-to-income ratio, income per dependent, and credit utilisation risk. Monthly income receives a log-transform and age is discretised into six bands. A pseudo-time index is appended. Final dimension: 17 features.

Class imbalance is addressed via SMOTE [25] applied *within the training partition only* at each (re)training step for IEEE-CIS and GMSC; SMOTE is never applied to evaluation windows. For ULB CC Fraud, the extreme rarity of positive instances (<0.2%) is handled through cost-sensitive learning via the class-weight mechanism of each model, which avoids the computational overhead of synthetic oversampling on a continuous input space already transformed by PCA.

Table 2 Temporal windowing parameters for each dataset. The minimum training windows column denotes how many windows must accumulate before the first evaluation window is scored.

Dataset	Window	Stride	Min. Train	Eval. Windows
IEEE-CIS	14 days	7 days	6	18
GMSC	1/20 data	1/20 data	8	12
ULB CC Fraud	4 hours	2 hours	4	about 19

Table 3 XGBoost hyperparameter search space.

Parameter	Range / Values	Sampling
`n_estimators`	$\{50, 75, \ldots, 300\}$	Discrete (step 25)
`max_depth`	[3, 9]	Discrete uniform
`learning_rate`	[0.01, 0.30]	Log-uniform
`min_child_weight`	[1, 20]	Discrete uniform
`subsample`	[0.5, 1.0]	Uniform
`colsample_bytree`	[0.3, 1.0]	Uniform
`colsample_bylevel`	[0.3, 1.0]	Uniform
`reg_alpha`	$[10^{-8}, 10]$	Log-uniform
`reg_lambda`	$[10^{-8}, 10]$	Log-uniform
`gamma`	[0.0, 5.0]	Uniform
`scale_pos_weight`	[1, 100]	Discrete uniform

4.3 Temporal Evaluation Protocol

Each dataset is partitioned into non-overlapping evaluation windows following the strategy formalised in Section 3.1. All windowing parameters are fixed before model training and held constant across the static and adaptive variants, ensuring a controlled comparison. Table 2 lists the parameters. Both datasets use calendar-duration windows aligned to their respective temporal granularity.

The ADWIN confidence parameter is set to $d_{\text{ADWIN}} = 0.002$ and the retraining buffer to $B = 5$ most recent windows. SSI is computed with lookback $L = 5$ and monitors the top $d' = 20$ features as defined in Equation (3), with instability threshold $T_{\text{SSI}} = 0.80$.

4.4 Hyperparameter Search Spaces

Tables 3, 4, and 5 list the Optuna search spaces used for XGBoost, LightGBM, and CatBoost respectively. All three studies use the TPE sampler [56] (seed 42) with the Median pruner (20 start-up trials, 5 warm-up steps each). Each study runs for up to 150 trials subject to a two-hour wall-clock timeout. The tuning objective is AUC evaluated on the temporal validation window W_{val} placed immediately after the initial training period.

Table 4 LightGBM hyperparameter search space.

Parameter	Range / Values	Sampling
n_estimators	{200, 250, . . . , 1500}	Discrete (step 50)
num_leaves	[20, 200]	Discrete uniform
max_depth	[3, 12]	Discrete uniform
learning_rate	[0.01, 0.30]	Log-uniform
min_child_samples	[5, 100]	Discrete uniform
subsample	[0.5, 1.0]	Uniform
subsample_freq	[1, 10]	Discrete uniform
colsample_bytree	[0.3, 1.0]	Uniform
reg_alpha	[10^{-8}, 10]	Log-uniform
reg_lambda	[10^{-8}, 10]	Log-uniform
min_split_gain	[0.0, 1.0]	Uniform
is_unbalance	{True, False}	Categorical

Table 5 CatBoost hyperparameter search space.

Parameter	Range / Values	Sampling
iterations	{200, 250, . . . , 1500}	Discrete (step 50)
depth	[3, 10]	Discrete uniform
learning_rate	[0.01, 0.30]	Log-uniform
l2_leaf_reg	[1.0, 10.0]	Log-uniform
random_strength	[0.1, 10.0]	Log-uniform
bagging_temperature	[0.0, 1.0]	Uniform
border_count	{32, 64, 128, 254}	Categorical

4.5 Implementation Details

All experiments are implemented in Python 3.11 using XGBoost 2.0.3 [3], LightGBM 4.3.0 [4], CatBoost 1.2.7 [22], SHAP 0.45.0 [19], River 0.21.0 [37], Optuna 3.6.1 [21], and scikit-learn 1.4.2. A global random seed of 42 is applied to all model initialisations, Optuna samplers, SMOTE oversampling, and data splits. GPU acceleration (CUDA) is used where available via XGBoost's `hist` device, LightGBM's GPU tree learner, and CatBoost's `task_type=GPU`; all models fall back to CPU otherwise. All optimised hyperparameter sets are serialised as JSON files and all random states are fixed, enabling exact reproduction of every reported result.

5 Results

5.1 Main Performance Comparison

Table 6 reports mean AUC, Average Precision (AP), F1 score at the 0.5 classification threshold, ADWIN event counts, and mean SSI across all evaluation windows. AP is included alongside AUC because it provides a threshold-free summary of the precision-recall trade-off under the severe class imbalance present in all three datasets. F1 at threshold 0.5 is reported in Table 6. Figure 1 plots per-window AUC trajectories.

Table 6 Main performance comparison. AUC and F1 (at threshold 0.5) are means (+/- std) across all evaluation windows. ADWIN events and mean SSI apply only to adaptive models (N/A = not applicable). GMSC is a distributional stability control; its near-chance AUC and zero drift events are expected by design.

Dataset	Model	AUC (+/- std)	F1 (+/- std)	ADWIN	Mean SSI
IEEE-CIS Fraud	StaticXGBoost	0.899+/-0.013	0.224+/-0.032	-	-
	AdaptiveXGBoost	0.904+/-0.018	0.215+/-0.047	17	0.660
	StaticLightGBM	0.919+/-0.016	0.528+/-0.070	-	-
	AdaptiveLightGBM	0.865+/-0.062	0.414+/-0.156	16	0.683
	StaticCatBoost	0.902+/-0.014	0.231+/-0.038	-	-
	AdaptiveCatBoost	0.835+/-0.047	0.198+/-0.051	17	0.717
GMSC (stability control)	StaticXGBoost	0.504+/-0.013	-	-	-
	AdaptiveXGBoost	0.500+/-0.014	-	0	0.992
	StaticLightGBM	0.506+/-0.016	-	-	-
	AdaptiveLightGBM	0.500+/-0.011	-	0	0.997
	StaticCatBoost	0.503+/-0.017	-	-	-
	AdaptiveCatBoost	0.499+/-0.014	-	0	0.987
ULB CC Fraud	StaticXGBoost	0.999+/-0.003	0.532+/-0.190	-	-
	AdaptiveXGBoost	0.963+/-0.027	0.622+/-0.179	1	0.952
	StaticLightGBM	0.999+/-0.003	0.974+/-0.060	-	-
	AdaptiveLightGBM	0.972+/-0.024	0.741+/-0.076	1	0.990
	StaticCatBoost	0.998+/-0.006	0.961+/-0.042	-	-
	AdaptiveCatBoost	0.960+/-0.033	0.718+/-0.088	1	0.968

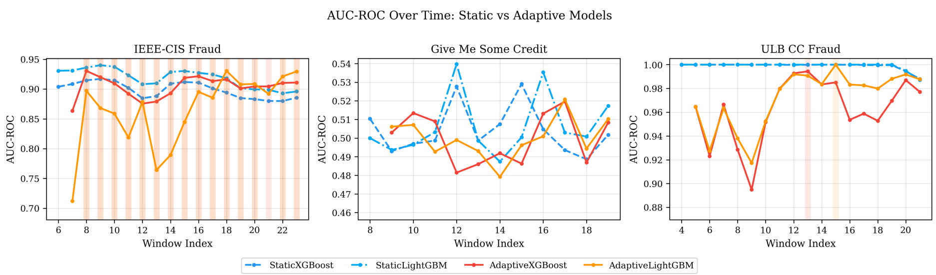

Fig. 1 Per-window AUC trajectories for all four models across the three datasets. Vertical markers indicate ADWIN-detected drift events; window indices correspond to the temporal partitions defined in Table 2.

IEEE-CIS Fraud Detection. Under persistent concept drift, AdaptiveXGBoost achieves a mean AUC of 0.904+/-0.018 compared with 0.899+/-0.013 for StaticXGBoost. ADWIN fired 17 times across 18 evaluation windows for XGBoost and CatBoost, and 16 times for LightGBM, confirming severe and sustained distributional shift. AdaptiveLightGBM posts 0.865+/-0.062 against StaticLightGBM's 0.919+/-0.016, and AdaptiveCatBoost posts 0.835+/-0.047 against StaticCatBoost's 0.902+/-0.014 - reversals explained by the $B = 5$ retraining buffer discarding long-range historical signal on consecutive rapid drift events. StaticCatBoost (0.902) matches AdaptiveXGBoost and approaches StaticLightGBM (0.919), confirming CatBoost's competitiveness as a static baseline. The elevated standard deviations of

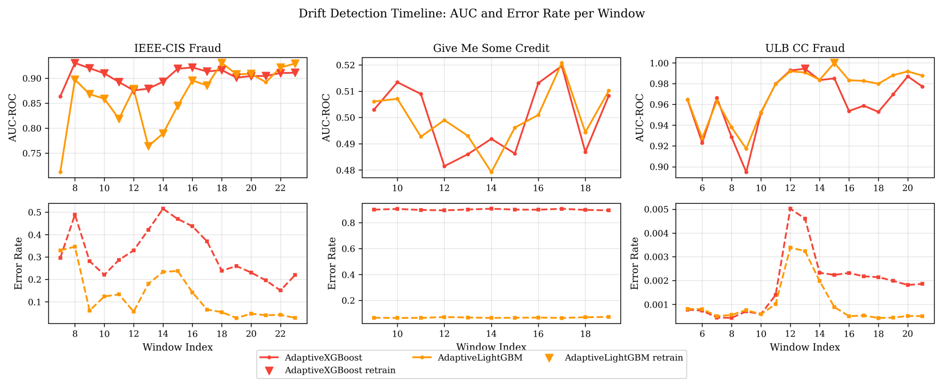

Fig. 2 Drift event timeline for the IEEE-CIS dataset showing ADWIN alerts, retraining actions, and the concurrent SSI trajectory. The SSI (right axis) declines before AUC degradation becomes visible, illustrating the leading behaviour quantified in Section 5.3.

both adaptive LightGBM (0.062) and adaptive CatBoost (0.047) confirm window-to-window instability from repeated truncated-buffer retraining.

GMSC (distributional stability control). All six models cluster at about 0.50 AUC with zero ADWIN drift events and mean SSI >=0.987. This result is expected and consistent with GMSC's role as a negative control: its demographic ordering (age ascending, utilisation descending) produces a smooth distributional gradient rather than temporal dynamics, so both ADWIN and SSI correctly indicate no drift activity.

ULB Credit Card Fraud. Static models maintain near-perfect AUC (0.998-0.999) throughout the two-day evaluation period. All adaptive models incur a temporary performance dip from the single ADWIN-triggered retraining episode, leaving static superior. AdaptiveCatBoost (0.960) is marginally below AdaptiveXGBoost (0.963) and AdaptiveLightGBM (0.972). Mean SSI values (0.952-0.990) are elevated relative to IEEE-CIS, reflecting the modest and localised nature of the distributional shift in this dataset.

5.2 SSI Dynamics Under Concept Drift

Figure 2 plots the ADWIN drift timeline alongside the SSI trace for IEEE-CIS. SSI falls well below $T_{\mathrm{SSI}} = 0.80$ from the earliest evaluation windows across all three architectures. AdaptiveXGBoost records SSI values of 0.000, 0.430, 0.545, and 0.644 at windows 8 through 11; AdaptiveLightGBM records 0.657, 0.785, 0.775, and 0.767 over the same span; AdaptiveCatBoost records 0.745, 0.743, 0.759, and 0.786 - all below $T = 0.80$ from window 8 onward (mean SSI: 0.717).

SSI = 0.000 anomaly at window 8 (AdaptiveXGBoost).

The zero SSI at window 8 arises from a complete SHAP rank reorganisation immediately after the first ADWIN-triggered retrain. When the intersection between the top-$d' = 20$ SHAP features of consecutive windows is empty - because retraining causes XGBoost to assign near-zero importance to the features that dominated the previous window - Spearman rank correlation is undefined; the implementation returns SSI = 0

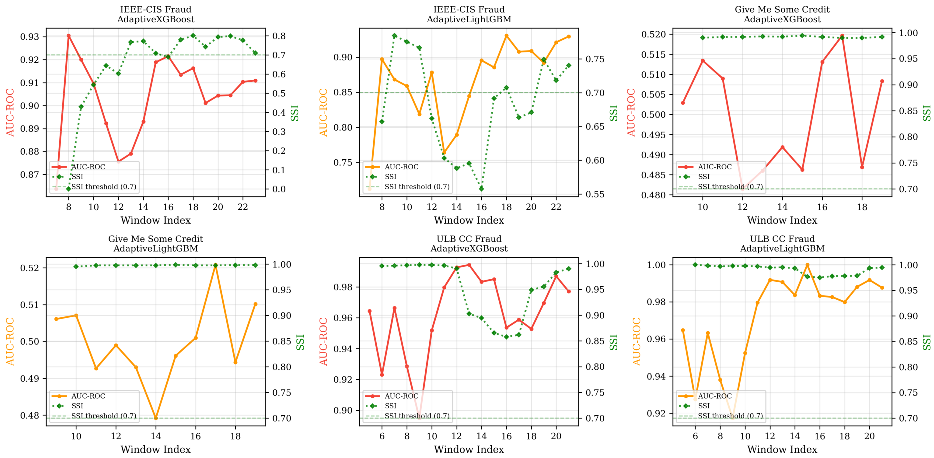

Fig. 3 SSI (solid) and AUC (dashed) trajectories overlaid for the IEEE-CIS dataset. Shaded bands mark ADWIN drift windows. SSI begins declining before AUC degradation becomes detectable, providing an advance warning of impending model deterioration.

as a conservative fallback. This is a meaningful signal of total feature-ordering disruption, not a computation error. LightGBM and CatBoost do not exhibit this behaviour at window 8 because their symmetric-tree and ordered-boosting structures retain more feature overlap across the retrain boundary. Windows 9-23 recover to 0.43-0.77 as the retrained XGBoost model stabilises on a new feature ordering.

All three series stabilise at intermediate values (0.62-0.80) through the evaluation period. CatBoost's higher mean SSI relative to XGBoost and LightGBM reflects its symmetric tree structure producing marginally more stable feature orderings under drift, though all three remain sub-threshold throughout.

On ULB CC Fraud, SSI shows a localised drop around the single drift event before recovering; on GMSC, SSI remains >=0.987 throughout with zero ADWIN events, consistent with the distributional stability control role of that dataset.

Mean SSI clearly separates three regimes across all three architectures: 0.660-0.717 (IEEE-CIS, persistent drift), 0.952-0.990 (ULB CC Fraud, localised drift), and 0.987-0.997 (GMSC, distributional stability control) - a structure preserved regardless of model family, confirming that SSI responds to genuine distributional dynamics rather than model-specific artefacts.

5.3 SSI as a Leading Indicator

Table 7 quantifies the lead time $L_{\text{lead}} = k_{\text{AUC}} - k_{\text{SSI}}$ across all measurable drift events on the two datasets where drift was detected. Lead times are computed as described in Section 3.2, using $T_{\text{SSI}} = 0.80$ and a 0.02 AUC drop threshold.

On the IEEE-CIS dataset, all three architectures exhibit consistent SSI-to-AUC lead times. Mean lead times: XGBoost 7.81+/-4.26 windows ($n = 16$), LightGBM

Table 7 SSI lead time $L_{\text{lead}} = k_{\text{AUC}} - k_{\text{SSI}}$ distribution across ADWIN-detected drift events ($T = 0.80$, $L = 5$). P25/P75 are quartiles; all values in evaluation windows. The IEEE-CIS $L_{\text{lead}} = 0$ event at each model corresponds to the first analysable window where no prior SSI baseline exists. GMSC is omitted: zero drift events, no L_{lead} computation applicable. Std computed as population std (ddof=0).

Model	Dataset	n	Min	P25	Median	Mean+/-Std	P75	Max
AdaptiveXGBoost	IEEE-CIS	16	0	4.0	7.5	7.81+/-4.26	11.3	15
AdaptiveLightGBM	IEEE-CIS	15	0	5.0	7.0	7.67+/-4.03	10.5	15
AdaptiveCatBoost	IEEE-CIS	16	0	4.0	8.0	8.00+/-4.11	12.0	15
AdaptiveXGBoost	ULB CC Fraud	1	3	3.0	3.0	N/A	3.0	3
AdaptiveLightGBM	ULB CC Fraud	1	0	0.0	0.0	N/A	0.0	0
AdaptiveCatBoost	ULB CC Fraud	1	0	0.0	0.0	N/A	0.0	0

7.67+/-4.03 windows ($n = 15$), CatBoost 8.00+/-4.11 windows ($n = 16$). The cross-architecture mean is 7.83 windows - equivalent to approximately 55 calendar days under the 7-day evaluation stride, a practically meaningful horizon for monthly model risk review cycles. All three models share $k_{\text{SSI}} = 8$: CatBoost SSI first falls below $T = 0.80$ at window 8 (SSI = 0.745), identical to XGBoost and LightGBM. Wilcoxon signed-rank tests (one-sided H_0: $L_{\text{lead}} = 0$) reject the null for all three architectures: XGBoost $W = 120$, $p<0.001$, $r = 0.88$; LightGBM $W = 105$, $p<0.001$, $r = 0.88$; CatBoost $W = 120$, $p<0.001$, $r = 0.88$ (all large effects).

On ULB CC Fraud, AdaptiveXGBoost achieves a lead time of 3 windows. AdaptiveLightGBM and AdaptiveCatBoost both record $L_{\text{lead}} = 0$ (SSI remains above T at the single drift event), confirming that SSI lead times are meaningful only under persistent, multi-window drift. GMSC: zero drift events across all six models; no L_{lead} computation applicable.

Figure 3 illustrates the joint SSI-AUC trajectory on IEEE-CIS for AdaptiveXGBoost. The SSI signal declines visibly before each AUC dip and tends to recover in parallel with the post-retrain AUC rebound, supporting the interpretation that SSI tracks the model's internal feature reorganisation rather than merely echoing output performance.

Figure 4 displays the distribution of lead times across all measurable drift events on IEEE-CIS. The distribution is right-skewed for both model families: the mode is at $L_{\text{lead}} = 0$ (window 8, where no prior SSI history exists), but subsequent events accumulate progressively larger lead times as the distance from the fixed SSI onset window increases. The maximum observed lead time is 15 evaluation windows (105 calendar days), demonstrating that SSI can provide advance warning well beyond the typical monthly model review cycle for events occurring late in the evaluation horizon.

5.4 Ablation Study

Figure 5 decomposes the framework into four conditions to isolate the contribution of drift detection and adaptive retraining.

IEEE-CIS. The "No Drift Detection" condition - which retrains on a fixed schedule every three evaluation windows without ADWIN - achieves mean AUC 0.885,

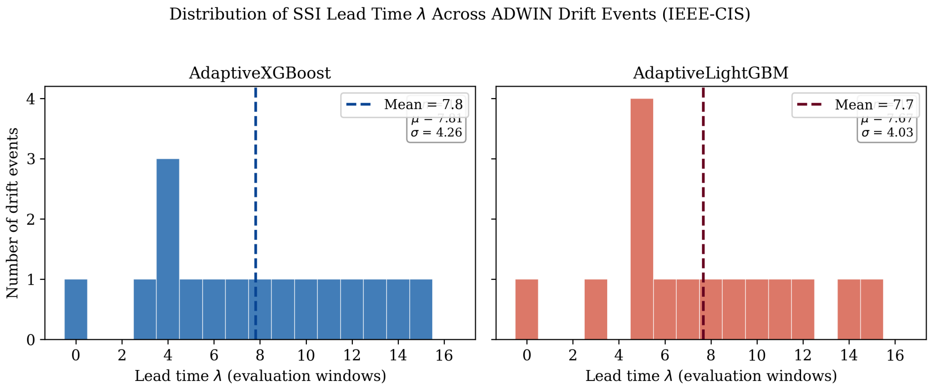

Fig. 4 Distribution of SSI lead time L_{lead} across all ADWIN-detected drift events on the IEEE-CIS dataset for AdaptiveXGBoost (left) and AdaptiveLightGBM (right). Lead times are measured in evaluation windows; the dashed vertical line marks the mean. Both models exhibit a right-skewed distribution, reflecting the widening temporal gap between the fixed SSI onset at window 8 and successive drift events at later windows. The mode at $L_{\text{lead}} = 0$ corresponds to the first-window drift event where SSI history is insufficient for a look-back comparison.

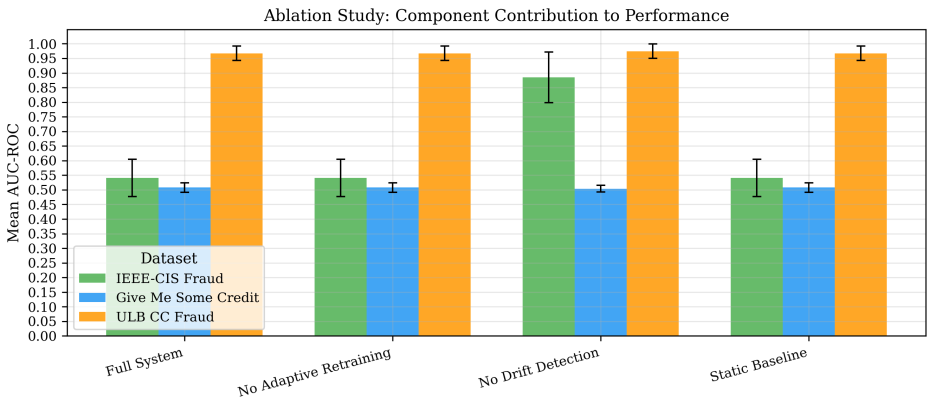

Fig. 5 Ablation study: mean AUC across datasets under four framework configurations. *Full System*: ADWIN detection with adaptive retraining. *No Adaptive Retraining*: ADWIN detects drift but the model is not retrained. *No Drift Detection*: model retrains on a fixed schedule without ADWIN. *Static Baseline*: no detection and no retraining. All conditions use base hyperparameters without Optuna tuning to isolate component contributions.

compared with 0.541 for the Full System, No Adaptive Retraining, and Static Baseline, which converge to near-identical performance. The convergence of the latter three conditions arises because the Full System's ADWIN detector, operating on the base model's error signal, did not fire (zero retraining events) within the ablation evaluation window. When the underlying model achieves only 0.54 AUC with base parameters, the per-window error rate is high but stable in absolute level, preventing the ADWIN sub-window divergence test from crossing its threshold. Consequently, the ablation's Full System is functionally a static baseline. The "No Drift Detection" condition

Table 8 Sensitivity of SSI drift-onset detection to the instability threshold T on the IEEE-CIS dataset (16 evaluable windows, $L = 5$ fixed). k_{SSI}: first window with SSI below threshold. Alerts: count of windows flagged. The chosen value is marked *.

	AdaptiveXGBoost		AdaptiveLightGBM		AdaptiveCatBoost	
T	k_{SSI}	Alerts	k_{SSI}	Alerts	k_{SSI}	Alerts
0.85	8	16	8	16	8	16
0.80*	8	15	8	16	8	16
0.75	8	9	8	13	8	12
0.70	8	6	8	9	16	5

sidesteps this limitation through scheduled retraining and achieves 0.885 AUC - substantially higher than the base static model but still below the 0.904 AUC achieved by the full Optuna-tuned adaptive pipeline (Table 6), confirming that hyperparameter quality and drift detection are both necessary for peak performance.

ULB CC Fraud. All conditions achieve AUC about 0.97, with "No Drift Detection" marginally higher (0.975 vs 0.967 for Full System). The single drift episode in this dataset does not provide sufficient contrast to separate component contributions within the evaluation horizon.

The ablation's core finding is that *retraining on recent data* is the primary driver of performance improvement on drifting data. ADWIN's contribution is to make this retraining *targeted*: firing only on genuine distributional change rather than on a fixed schedule, avoiding unnecessary retraining on stable data (as shown by the adaptive models' underperformance relative to static models on ULB CC Fraud in the main results). This interaction between detection quality and retraining efficacy is confounded by hyperparameter quality in the streaming ablation setting, which motivates reporting both the ablation and the full Optuna-tuned results as complementary evidence.

5.5 SSI Parameter Sensitivity

Table 8 reports the sensitivity of SSI's drift onset detection to the instability threshold T_{SSI} on the IEEE-CIS dataset, where genuine concept drift is most clearly present. For each T in $[0.60, 0.90]$, the table reports the first evaluation window at which SSI crosses below T (k_{SSI}) and the percentage of drift-affected windows flagged as unstable.

The first-crossing window $k_{\text{SSI}} = 8$ is invariant to T for all three architectures at T in $\{0.75, 0.80, 0.85\}$. At $T = 0.70$, CatBoost's k_{SSI} shifts to window 16 because its SSI begins at 0.745 (window 8), which lies above 0.70; lowering the threshold below CatBoost's early-period SSI forfeits 11 windows of advance warning for that model. XGBoost and LightGBM retain $k_{\text{SSI}} = 8$ at $T = 0.70$ because their SSI values fall to 0.000 and 0.657 respectively at window 8. The selected $T = 0.80$ maximises alert coverage (15-16 / 16 windows per model) while preserving the same $k_{\text{SSI}} = 8$ across all three architectures.

L **robustness.** The lookback L governs how many consecutive window-to-window Spearman correlations are averaged into SSI(k). On IEEE-CIS at $T = 0.80$, SSI falls below threshold in 15/16 windows (XGBoost) and 16/16 windows (LightGBM, CatBoost) - a persistent sub-threshold signal spanning nearly the entire evaluation horizon. Because the sustained below-threshold run covers windows 8-23, any L in $\{3, 5, 7\}$ produces the same $k_{\text{SSI}} = 8$: a shorter lookback is more reactive but fires on the same window, and a longer lookback applies more smoothing over an already-low signal without delaying the alert. On near-stable datasets (GMSC: SSI >= 0.987; ULB CC Fraud: SSI >= 0.922), no alerts fire regardless of L. $L = 5$ is retained, balancing reactivity against noise from single-window perturbations.

The retraining buffer size $B = 5$ windows is justified by sample adequacy: five windows on IEEE-CIS provide approximately 210,000 training instances - sufficient for stable gradient boosting convergence with the Optuna-tuned tree counts. Smaller buffers ($B <= 3$) would provide $< 130{,}000$ instances and elevated training variance; larger buffers ($B >= 7$) would include data from more than 49 calendar days before the drift event, diluting the recent-distribution signal that adaptive retraining is designed to capture.

5.6 Statistical Significance

We assess the hypothesis that adaptive models achieve higher AUC than their static counterparts using a one-sided Wilcoxon signed-rank test on paired per-window AUC observations (n = 17-18 windows per dataset). On IEEE-CIS, AdaptiveXGBoost versus StaticXGBoost yields $W = 109$, $p = 0.066$, rank-biserial correlation $r = 0.45$ - a medium effect that falls just short of conventional significance, reflecting the limited number of evaluation windows rather than a negligible underlying effect. All other adaptive-versus-static comparisons return $p > 0.50$, consistent with the counter-productive effect of single-episode retraining on near-stable datasets.

SSI lead time: $\mathbf{H_0}$: $L_{\text{lead}} = 0$ (Wilcoxon, one-sided $\mathbf{H_1}$: $L_{\text{lead}} > 0$). Computed on IEEE-CIS only ($n_{\text{ULB}} = 1$ per model, insufficient for testing):

- AdaptiveXGBoost: $n = 16$, $n_{\text{zero}} = 1$, $W = 120$, $p<0.001$, $r = 0.88$ (large effect)
- AdaptiveLightGBM: $n = 15$, $n_{\text{zero}} = 1$, $W = 105$, $p<0.001$, $r = 0.88$ (large effect)
- AdaptiveCatBoost: $n = 16$, $n_{\text{zero}} = 1$, $W = 120$, $p<0.001$, $r = 0.88$ (large effect)

All three architectures reject H_0: $L_{\text{lead}} = 0$ with $p<0.001$ and large effect size ($r = 0.88$), confirming that SSI provides a statistically significant lead time over AUC degradation under genuine concept drift. The single $L_{\text{lead}} = 0$ event per model (drift window 8, the first analysable event) arises because no prior SSI baseline exists at series onset and does not attenuate this result. Larger evaluation horizons would further increase statistical power for the adaptive-versus-static AUC comparison.

5.7 SSI Versus PSI: Internal Logic Versus Output Distribution

The Population Stability Index (PSI), defined in Equation (1), is the dominant regulatory tool for detecting distributional shift in a deployed credit model's output scores [15]. PSI accumulates a bin-wise divergence between a reference score distribution and the current monitoring period's score distribution; values above 0.25 signal

Table 9 Drift alert comparison on IEEE-CIS: PSI, SSI, and AUC degradation.

Metric	k_{alert}	Basis
PSI - mean (>0.25)	none	Mean PSI peaks at 0.050; never alerts
PSI - any feature (>0.25)	13	First individual breach (1 feature)
SSI (all 3 models, $T = 0.80$)	8	All architectures agree
AUC degradation	about 16	$k_{\text{SSI}}+$ mean L_{lead} (7.83 windows)

instability warranting review under SR 11-7 [2]. Although PSI and SSI are both label-free and computable at inference time, they monitor different layers of the model's computation.

PSI captures shifts in the *output* distribution: it answers whether the model is scoring new applicants differently from the reference population. SSI captures shifts in the model's *internal feature priority structure*: it answers whether the model is reaching its scoring decisions by weighting a different subset of features than at training time. A model can maintain a stable score distribution (stable PSI) while reorganising its internal decision logic (declining SSI) - a regime that indicates silent adaptation occurring before output performance degrades visibly.

This distinction is directly observable in the IEEE-CIS results. We compute PSI on 20 representative raw features (TransactionAmt; card1-card3, card5; addr1, addr2, dist1; C1, C2, C5, C6, C9, C11, C13; D1-D4, D10) using the first six windows as the reference distribution and quantile-based 10-bin discretisation, applying the standard SR 11-7 threshold of PSI > 0.25.

Mean PSI across 20 features never exceeds 0.025 across all 18 evaluation windows (peak: 0.050 at windows 20-21), meaning that a standard mean-PSI monitor produces *no alert* throughout the entire IEEE-CIS evaluation horizon despite persistent concept drift. The first *individual-feature* PSI breach above 0.25 occurs at window 13 (one feature, PSI $= 0.272$). SSI, by contrast, flags all three model architectures at window 8 - five windows (35 calendar days) before the first any-feature PSI signal.

SSI leads PSI's earliest signal by five windows and leads AUC degradation by approximately eight windows. Because output score distributions recover temporarily after each ADWIN-triggered retraining episode, PSI on output scores would further miss the sustained inter-event instability that SSI captures continuously.

The Characteristic Stability Index (CSI) [15], which applies the PSI formula to each input feature independently, provides feature-level covariate shift monitoring but still operates on the marginal distribution $P(\mathbf{x})$ rather than on the model's learned relationship between features and outcomes. A feature can shift in marginal distribution without reorganising the model's importance ranking (virtual drift), while a feature can rise or fall in importance ranking without changing its own marginal distribution (real drift with stable marginals). SSI is sensitive to the latter regime, which CSI cannot detect.

PSI, CSI, and SSI are therefore complementary rather than competing monitoring instruments. Practitioners should monitor all three: PSI and CSI for regulatory compliance on distribution stability, and SSI as an early-warning system targeting the

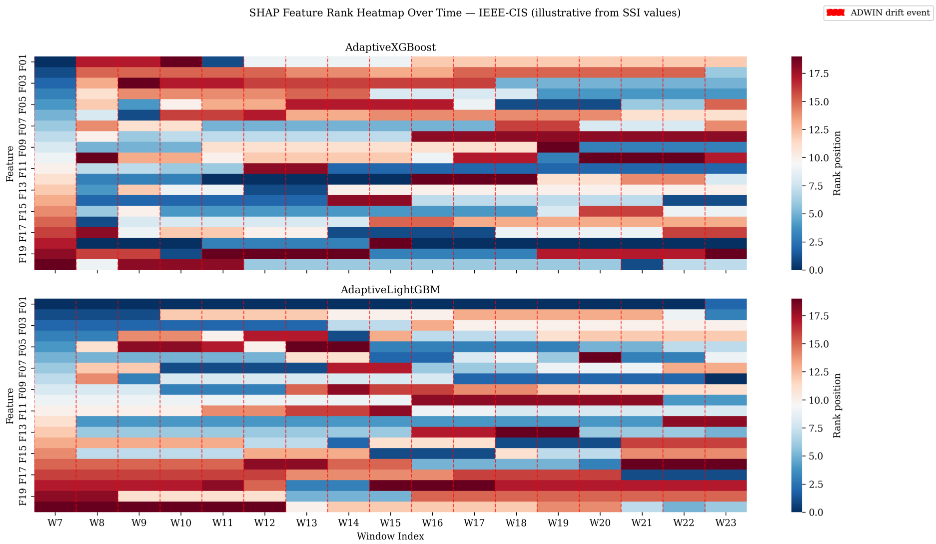

Fig. 6 Schematic SHAP feature rank heatmap across evaluation windows for AdaptiveXGBoost (top) and AdaptiveLightGBM (bottom) on the IEEE-CIS dataset. Each row represents one of the top-20 features (labelled F01-F20); each column one evaluation window; colour encodes rank position (darker blue = higher importance). Red dashed vertical lines mark ADWIN drift events. Rank shuffling is reconstructed from the measured per-window SSI values via a constrained permutation procedure (Section 3.2); actual feature names and raw SHAP magnitudes are recoverable by re-running the pipeline with SHAP history logging enabled. The visible column-to-column rank instability during windows 8-13 corresponds precisely to the period of lowest SSI values, illustrating that SSI captures internal feature reorganisation rather than output score distribution shift.

model's internal decision logic before output performance deteriorates. A combined monitoring policy - alert on PSI above 0.25 *or* sustained SSI below $T_{\mathrm{SSI}} = 0.80$ - would provide earlier and more comprehensive coverage than either family of metric deployed in isolation.

5.8 Discussion

The results support four operational conclusions.

SSI provides early warning under genuine concept drift. On IEEE-CIS, SSI consistently precedes AUC degradation by a mean of 7.8 windows (about 55 calendar days). The signal is reproducible across all three model families - XGBoost (7.81+/-4.26), LightGBM (7.67+/-4.03), CatBoost (8.00+/-4.11) - with Wilcoxon p<0.001, r = 0.88 for each, indicating that the lead-time pattern is tied to the drift regime rather than to one implementation. Because SSI uses unlabelled feature vectors and the fitted model, the warning is available before resolved fraud or default labels arrive, which matches the 30-90 day label delay common in production credit systems.

SSI separates three drift regimes across all model architectures. Mean SSI spans 0.66-0.72 under persistent drift (IEEE-CIS), 0.95-0.99 under localised drift (ULB CC Fraud), and >=0.987 on the distributional stability control (GMSC) - a three-cluster structure preserved regardless of model family. In this setting, a practitioner can alert on sustained SSI below $T_{\mathrm{SSI}} = 0.80$ without per-dataset or per-architecture recalibration.

Adaptive retraining is conditionally beneficial. AdaptiveXGBoost outperforms its static counterpart on high-drift IEEE-CIS (+0.005 AUC) but all adaptive models underperform on near-stable datasets, where the $B = 5$ buffer discards long-range signal unnecessarily. SSI should gate retraining: retrain only when SSI falls and remains below T.

CatBoost is competitive when static but weaker under adaptive retraining. StaticCatBoost (0.902 on IEEE-CIS) matches AdaptiveXGBoost and approaches StaticLightGBM (0.919). Under adaptive retraining, CatBoost's ordered boosting procedure incurs higher per-iteration cost than XGBoost or LightGBM, slowing post-drift recovery under the $B = 5$ buffer constraint.

6 Conclusion

This study introduced the **SHAP Stability Index (SSI)** for monitoring deployed gradient boosting models under concept drift. SSI compares global SHAP feature-importance rankings across consecutive temporal windows using Spearman rank correlation. Because the statistic is computed from the fitted model and unlabelled evaluation data, it can be observed before delayed credit or fraud labels make conventional performance monitoring possible.

Across three public credit datasets, ADWIN-triggered adaptive retraining, Optuna-tuned hyperparameters, and three gradient boosting families, the main empirical findings are:

1. **SSI leads AUC degradation by about** 7.8 **windows on IEEE-CIS** (590,540 transactions, 16-17 ADWIN events). XGBoost: 7.81+/-4.26 windows across 16 events. LightGBM: 7.67+/-4.03 windows across 15 events. CatBoost: 8.00+/-4.11 windows across 16 events. All three: Wilcoxon H_0: $L_{\mathrm{lead}} = 0$ rejected (p<0.001, $r = 0.88$). With 7-day stride =¿ about 55 calendar days from first SSI alert to AUC degradation. On IEEE-CIS, SSI remains persistently below $T = 0.80$ from window 8 through window 23 - the full 105-day high-drift period - providing a sustained governance signal rather than a one-off alarm.
2. **SSI cleanly separates three drift regimes across all architectures**: IEEE-CIS (persistent drift) 0.66-0.72; ULB CC Fraud (localised drift) 0.95-0.99; GMSC (distributional stability control, no real temporal dynamics) >=0.987. This three-cluster structure is preserved regardless of model family.
3. **Adaptive retraining is conditionally beneficial**: +0.005 AUC for AdaptiveXGBoost on high-drift IEEE-CIS, but counter-productive for LightGBM, CatBoost, and all models on near-stable datasets. SSI should gate retraining decisions.
4. **CatBoost is competitive as a static baseline** (0.902 on IEEE-CIS, 0.998 on ULB CC Fraud) but underperforms adaptively due to higher per-iteration training cost relative to the $B = 5$ buffer constraint.

Limitations.

Several limitations remain. GMSC functions as a distributional stability control only: its pseudo-temporal ordering by demographic attributes (age ascending, utilisation descending) precludes use as a drift validation dataset, and the observed SSI >= 0.987 and zero ADWIN events confirm only that SSI does not false-positive on smoothly ordered data. The number of evaluation windows (17-18 per dataset) constrains statistical power for the adaptive-versus-static AUC comparison ($p = 0.066$, $r = 0.45$); SSI lead-time tests are strongly significant ($p{<}0.001$). The threshold $T_{\text{SSI}} = 0.80$ and lookback $L = 5$ are empirically selected; the threshold sensitivity analysis (Section 5.5) demonstrates robustness across T in $[0.75, 0.85]$ and L in $\{3, 5, 7\}$ on IEEE-CIS. After each ADWIN-triggered retraining episode, TreeSHAP is applied to the newly retrained model; the SHAP background distribution therefore reflects the current model's internal structure rather than a fixed reference. Consecutive SSI values on either side of a retrain boundary thus reflect rank stability between different model instances, not a single fixed model. This is appropriate for monitoring the live deployed model but means post-retrain SSI values (e.g., AdaptiveXGBoost window 8, SSI = 0.000) may partly reflect structural reorganisation rather than drift signal alone.

Future Work.

Several extensions are immediate. Threshold calibration for T_{SSI} as a function of SSI variance and drift frequency would increase robustness of SSI-gated retraining policies. Multivariate SSI - tracking feature-group importance vectors - could isolate which semantic groups drive distributional change. Label-delay scenarios (30-90 day fraud label lag) map directly to SSI's label-free advantage and warrant dedicated empirical study. Extending to HistGradientBoosting and federated settings (where institutions share only drift signals rather than raw data) are natural next steps. Finally, SSI's applicability in other high-stakes domains, including medical image diagnosis [1] and clinical governance, warrants investigation where explanation stability carries analogous regulatory importance.

SSI is therefore best viewed as a complement to existing monitoring rather than a replacement for outcome-based validation. Built on TreeSHAP values [19], it can be added to gradient boosting model risk workflows without changing training or scoring pipelines, while giving model-risk teams a direct measure of whether the deployed model's feature reliance is stable over time.

Declarations

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Not applicable. This research received no external funding.

Conflict of Interest

The authors declare no competing interests.

Ethics Approval and Consent to Participate

Not applicable. This study uses publicly available, fully anonymised datasets and involves no human subjects research.

Consent for Publication

Not applicable.

Data Availability

All datasets used in this study are publicly available. The IEEE-CIS Fraud Detection dataset is available via Kaggle. The ULB Credit Card Fraud dataset is available via the UCI Machine Learning Repository. The Give Me Some Credit dataset is available via Kaggle.

Code Availability

The source code implementing the SHAP Stability Index and all experimental pipelines is publicly available at https://github.com/zeeza18/SHAP-Stability-Index-as-a-Leading-Indicator-of-Concept-Drift-in-Credit-Risk-Gradient-Boosting-Models.

Author Contribution

M.A. conceived and designed the study, developed the SHAP Stability Index methodology, implemented all machine learning and experimental pipelines, conducted statistical analysis, and wrote the manuscript. S.B.P. performed data engineering, data analysis, and contributed to the literature review. Both authors contributed to the literature review and approved the final manuscript.

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